# MD-DGFLS: MULTI-DOMAIN DEEPFAKE FORENSIC VERIFICATION CERTIFICATE

IEEE TPAMI / CVPR Technical Forensics Specification v2.2-PROD | Cryptographic Audit Dossier

| | | | |
|---|---|---|---|
| **File Name:** | 0b49747b_synth.png | **Target Domain:** | Face |
| **Audit Timestamp:** | 2026-09-25 10:26:19 UTC | **SHA-256 Hash:** | b7877434e79a37fbbe0cfdd4... |
| **Forensic Verdict:** | **VERIFIED AUTHENTIC MEDIA** | **Confidence:** | **58.4%** |
| **Tamper Type:** | Authentic Media Stream | **Bounding Box:** | None |

## Three-Stage Visual Forensics Pipeline



**Stage 1: Source Image**




**Stage 2: Tamper Overlay**



**Stage 3: XAI Heatmap**

## Mathematical Innovations & Multi-Expert Telemetry

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|---|---|
| **CDFAM (Cross-Domain Alignment)** | Contrastive Score: 0.4876 (Margin: m=0.3) |
| **ARCC (Entropy Calibration)** | Dynamic Calibration Temperature: T = 1.0 |
| **TGHFL (Transformer-GAN Cross-Attention)** | Cross-Attention Energy: 0.0005 |
| **MEFCN (Multi-Expert Consensus)** | Face: 71.1% \| Medical: 9.5% \| Satellite: 6.6% \| Regular: 6.4% \| Natural: 6.4% |
| **HLRN (Localization Refinement)** | Coarse-to-Fine Refinement: 256x256 Native |
| **Frequency Stream (ELA & Laplacian)** | ELA Discrepancy: 1.081 \| Sensor Noise Var: 1383.1 |

This certificate is cryptographically anchored and generated by the MD-DGFLS Autonomous Forensics Engine. Verified under IEEE TPAMI standards.